

How Is Time Produced?

Chujia Kuang (邝楚嘉)

嘉言一得 (Jiayan Yide)

Independent Researcher, Jieyang, Guangdong, China

2026-10-04 · 预印本 (preprint) · 未经同行评议 · v2

DOI: 10.5281/zenodo.22906313

Chujia Kuang (邝楚嘉)

Independent Researcher, Jieyang, Guangdong, China

Corresponding author: Chujia Kuang, Independent Researcher, Jieyang, Guangdong, China. E-mail: chirca@163.com

Abstract

A day is said to have twenty-four hours, and a watch keeps to that figure. The figure is not exact. One revolution of the earth, measured against the fixed stars, takes 23 hours 56 minutes 4 seconds; the missing four minutes are consumed by the earth's motion along its orbit, so that the sun returns overhead only after the earth has turned a little further. Both statements are true; they count different things. This note asks where time comes from, and answers that it was never produced: it is the reading a person records against several periodic motions. The second is defined by the vibration of a caesium-133 atom, because only a motion that repeats without end can serve as a ruler. Han texts already say as much: the *Shizi*, the *Zhuangzi* and the *Mozi* tie rest to time as firmly as motion. Three rulers — day, month and year — are then set against their motions; all three give non-integer readings, and the whole difficulty of the calendar grows from those remainders. The same motion, measured against the sun, the moon or the fixed stars, yields different figures: 365.2422 solar days to a tropical year, but 366.2422 sidereal days. The remainders are shown to have been counted on the ground, in the 1461-day cycle and the four-minute nightly shift of the stars. Finally, because the earth's rotation is slowing under tidal friction, the leap second is needed to hold civil time in step. The note is descriptive; it contains no predictive claims.

Keywords: time; periodic motion; sidereal day; solar day; tropical year; precession; leap second

“A day has twenty-four hours.”

Everyone says it, and the watch runs by it.

But it is not exact.

One full revolution of the earth — from this instant, turning back to the direction it faced before — takes 23 hours 56 minutes 4 seconds. Nearly four minutes short of twenty-four hours.

The four minutes are not lost. Another motion eats them: the earth turns, and at the same time it travels forward. By the time it has completed one turn, the sun is no longer in its old position, and the earth must turn a little further before the sun comes back overhead.

A little further each day, and there are the four minutes.

So “a day is twenty-four hours” and “one turn of the earth is 23 hours 56 minutes” are both correct; they are simply not counting the same thing.

One step further down: where does time come from?

1. Only periodic motion can serve as a ruler

We are used to thinking of time as a kind of thing, like the water of a river, flowing of itself, with the clock merely measuring how much has passed.

In its modern definition it is not that at all. The hardest example is the second.

In 1967 the Thirteenth General Conference on Weights and Measures redefined the second:

The second is the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom.

That definition is still in use. The 2018 revision of the International System of Units left the second built on this frequency, only written differently.

There is a detail here worth turning over: why choose the vibration of an atom, of all things, to define how long a second is?

Because only a motion that repeats, over and over, can serve as a ruler.

The moon circling the earth, the earth circling the sun, the earth turning on its own axis, the electron in an atom jumping back and forth between two levels — these are all periodic: after an interval they begin again. The day one of them stops repeating, it stops being usable as a ruler.

So “time” is not a thing that was discovered. It is the reading a person records against several periodic motions.

Which motions, then?

2. Two thousand years ago the Chinese were already saying this

Start with 宇 and 宙.

The people of the pre-Qin period joined space and time in a single word, 宇宙. The word is still in use today, but its original sense was far more substantial than the present one:

「天地四方曰宇，往古来今曰宙。」——《尸子》（“Heaven, earth and the four directions are called yu; what is past and what is to come are called zhou.” — Shizi）

宇 is heaven, earth and the four directions — that is space; 宙 is from antiquity to the present — that is time.

The Huainanzi, “Qisu xun”, puts it almost identically: 往古来今谓之宙，四方上下谓之宇（“what runs from antiquity to the present is called zhou; the four directions, above and below, are called yu”）.

The Zhuangzi, “Gengsang Chu”, takes another angle:

「有实而无乎处者，宇也；有长而无本剏者，宙也。」（“That which has reality but no fixed place is yu; that which has length but no beginning or end is zhou.”）

Space is solid and real, yet no fixed place can be found for it; time keeps growing, yet no head and no tail can be found for it.

(One caution: the original Shizi was long since lost, and the present text is a reconstruction by later hands; this line is cited in the commentary to the Shishuo xinyu, “Paitiao”, so in quoting it one must say so.)

The Mohists went a step further.

In the Mozi, “Jingshang”, there are two pairs of lines worth reading together:

「久，弥异时也。」「宇，弥异所也。」「动，或徙也。」「止，以久也。」（“Jiu [duration] is what covers the different times; yu is what covers the different places.” “Motion is the shifting of a region; rest is measured by jiu [duration].”）

The first pair says: 久, that is time, strings every different moment together; 宇 strings every different place together.

The second is more important. 或 is the ancient form of 域, so 「动，或徙也」 means: motion is the migration of a region of place. And 「止，以久也」 means: rest, too, must be measured by time (久).

People of more than two thousand years ago already saw it clearly: without time, not even “rest” can be stated. Motion and rest both hang on time at either end.

This is the same direction as present-day physics; only the ancients said it in five characters.

Take one piece of positive evidence. The *Shangshu*, “Yao dian” , records Yao's command:

「钦若昊天，历象日月星辰，敬授民时。」 (“Reverently follow Heaven above; observe and compute the sun, the moon and the stars; respectfully bestow the seasons on the people.”)

历象 is observing and computing; 敬授民时 is bestowing the computed seasons on the common people. The seasons do not drop from the sky; they must be read out of the motion of the sun, the moon and the stars, and then passed on. This sentence is, by and large, the governing principle of every calendar that followed.

3. Three rulers, all of them motion

The oldest units of time — not one of them was “made” by anyone; all were “counted” .

Unit	Which motion is counted	Reading
Day	One rotation of the earth (counted against the sun)	1 mean solar day = 24 hours
Month	One cycle of the moon's phases (counted against the phase)	1 synodic month $\approx$ 29.53 days
Year	One circuit of the earth around the sun (counted against the vernal equinox)	1 tropical year $\approx$ 365.2422 days

Three motions, three units. The motion comes first, the unit after; it is not that a unit came first and was then fitted to a motion.

But the three rulers share one defect: none of them counts out to a whole number.

A year is not exactly 365 days. A month is not exactly 29 and a half days. A day is not exactly 24 hours.

What the calendar has had to deal with from its first day is these remainders. The lichun, the solar terms and zhengyue treated in the earlier pieces all have their root here.

4. The same motion, a different reference, a different count

This is the most important part of the whole piece.

To count a motion one must first find something “not moving” to serve as reference. But a different reference gives a different count. The same motion can be reported as several

different times.

Unit	Counted against the sun or the phase	Counted against the fixed stars	Difference
Day	mean solar day = 24 hours	mean sidereal day = 23 hours 56 minutes 4 seconds	about 3 minutes 56 seconds
Month	synodic month = 29.53059 days	sidereal month = 27.32166 days	about 2.21 days
Year	tropical year = 365.24220 days	sidereal year = 365.25636 days	about 20 minutes 24 seconds

(For the last digits of the differences, each source has its own slight variation. Take the difference between the solar day and the sidereal day: some write 3 minutes 56 seconds, others 3 minutes 57 seconds.)



Why, counted against the sun, is the day longer? The reason in one sentence:

The earth turns, and at the same time travels forward.

By the time it has completed a full turn and returned to its former direction, the sun has moved forward along the orbit by about 1 degree. The earth must turn that much more before the sun comes back directly overhead.

So the solar day = one rotation + the little while needed to turn about 1 degree more.

Push this out over a whole year and a counter-intuitive number appears:

In one tropical year there are 365.2422 solar days, but 366.2422 sidereal days.

That is to say, in the course of this year the earth in fact turned more than 366 times; yet we lived through only a little more than 365 days.

The extra turn is not a miscalculation. It is eaten by the question “what is taken as the reference” : one day eats 3 minutes 56 seconds, and over 365 days that comes to exactly one whole turn.

This layer also settles the earlier question of “why there are three calendars” . None of the three calendars miscalculated; each simply took a different base: take the sun as base and you get the rhythm of the “year” ; take the fixed stars and you get the rhythm of the “day” ; take the phase and you get the rhythm of the “month” .

5. Those remainders were counted on the ground

The remainders were counted out by the ancients, day by day.

Take the year. Through long observation the ancients found that from the moment the sun's shadow returns to its former place to the moment it coincides with the previous one, the interval is 1461 days, exactly four years. Divide by four, and a year is 365.25 days.

Then what of the extra day? $1461 = 365 \times 4 + 1$; that surplus day is the leap day.

Why the leap day comes every four years — the answer is here, not in a formula.

Now the 3 minutes 56 seconds. A little each day, accumulated over 365 days, makes exactly one whole day, and that is the extra 366th turn. The account balances.

There is one more thing visible merely by looking up: the same star rises about four minutes earlier each night than the night before.

Four minutes a day is nothing to notice. A month makes two hours; over a year, the whole sky of stars has turned over once. That the star patterns of the four seasons change has its root in these 3 minutes 56 seconds.

6. Motion changes, and time changes with it

If time really is the reading of motion, there is a layer further down still: motion itself changes, and time should change with it.

This has already been measured in astronomy.

The earth's rotation is slowly slowing. The main cause is tidal friction. The moon's gravity raises the sea water, and the sun does its part as well, a smaller one; as the tide grinds across the ocean floor it carries friction, which is to say it puts a brake on the earth's rotation.

How much slower? About every hundred years, a day grows by nearly two milliseconds. (Values differ slightly from source to source, roughly between 1.7 and 2 milliseconds; this is an order of magnitude, not a figure fixed to the decimal place.)

Two milliseconds is imperceptible to a person today. Put it into geological time, though, and it is startling: by one study, a day in the age of the dinosaurs was only about 23.5 hours.

Hence the leap second.

The length of the second kept by the atomic clock is constant; the earth's rotation is not. The two run each on its own, and in time they separate. From 1972 the world agreed: when the gap grows large, a second is added to Coordinated Universal Time, holding the difference within 0.9 of a second.

From 1972 to the present, twenty-seven such seconds have been added, every one of them an addition, never once a subtraction. All of them “additions” shows precisely that the earth is slowing.

In 2022 the Twenty-seventh General Conference on Weights and Measures passed a resolution: no later than 2035, leap seconds will no longer be introduced.

There is a still slower layer: the direction of the motion itself is also drifting.

The earth's axis of rotation traces a cone in space, and the vernal equinox retreats westward year by year. The direction in which a given star lay two thousand years ago is no longer where it lies today. This is called precession. It is too slow; the earlier pieces have left it alone, and a piece of its own comes later.

Closing

So back to the question at the start: how is time produced?

It was never produced. It is the reading a person records against several motions.

One turn of the earth is a day; one cycle of the moon's phases is a month; one circuit of the earth around the sun is a year.

Motion first, graduations after. Motion was already there; the graduations are added by human beings.

The ancients recorded these readings, aligned them, cut them into segments, and so had the calendar. And the neat twenty-four hours on our watches today is in fact a ruler that has been smoothed — the true sun does not run evenly; it is the “mean sun” that smooths it on its behalf. The earlier piece on clock time and true solar time spoke of “one uneven, one even” ; this is what it meant. The sidereal day and the solar day here are another pair of differences within the same family.

What comes next is the several roads in the sky — the one the sun travels, the one the moon travels, and the one used to measure time.

Note by Chujia Kuang

The reason for taking up this subject is to bring the word “time” down to the ground. It is not a thing that can be accumulated; it is the reading a person records against several periodic motions. Once this layer is understood, the earlier pieces have their origin: why the start of spring is no more than a single instant, why the first month shifted several times, why the solar terms differ by a few hours from year to year — all those remainders grew from here.

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record (时间是怎么产生的?).

References

- [1] Shi Jiao. Shizi [尸子][M]. Pre-Qin period.
- [2] Zhuang Zhou. Zhuangzi [庄子][M]. Pre-Qin period.
- [3] Mo Di. Mozi [墨子][M]. Pre-Qin period.
- [4] Shangshu [尚书][M]. Pre-Qin period.
- [5] Liu An. Huainanzi [淮南子][M]. Western Han.
- [6] Liu Yiqing. Shishuo xinyu [世说新语][M]. Northern and Southern Dynasties.